Biopatternsg: A logical and ontological framework for knowledge discovery on gene regulatory networks.

Description, installation, execution, consultation methods and standard results.

The Researcher's Guide

(Version 3.0. Revision 04)

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Introduction

This project combines Generative AI and Logic AI to form an integrated process, a pipeline, that retrieves large volumes of scientific information and produces logical models that can be validated and leveraged by human experts. Our goals are to organize and to analyze, with the assistance of AI and other bio-information retrieval tools, the documentation associated with a domain and extract validated knowledge to support planning for problem solving. We have tested our methodology and the pipeline with an experiment on COVID-19 which we are also prepared to report. The tool helped a team of biologists to review the state of the art and handle the explosion of research papers during pandemia. The main product of the pipeline is a Gene Regulatory Network (GRN); a semantic map which constitutes a representation of causal relations in system biology which can in turn be used to explore different strategies for intervention and treatment. In a GRN, a regulatory event can activate a product that participates in an event, which activates or inhibits another product. Thus a GRN can be used to answer questions like: given a drug, what kinds of protein-protein interactions result when that drug

binds to a known receptor, and which of them lead to the activation (or inhibition) of the transcriptional response of some gene, or the up/down regulation of a disease?. The pipeline (named Biopatternsg) has been structured around three main tasks: a) data and information gathering, from bioinformatics portals such as MeSH (ncbi.nlm.nih.gov/mesh), and GeneOntology (geneontology.org); b) the use of generative AI and Large Language Models (LLMs) for the automatic extraction of biological entities and their biological relationships, based on the services of PubTator (ww.ncbi.nlm.nih.gov/research/pubtator3); and c) the use of logic AI to manage the information obtained in a) and b) and combine it with conserved criteria, all represented as knowledge bases in Prolog (http://www.swi-prolog.org) and constraint logic programming. The system also keeps track of and allows access to all the original documents used as sources of knowledge. It is possible, for instance, to choose an inferred regulatory pathway and retrieve the abstracts that support it. A biological analysis conducted by human experts has concluded that the pipeline is useful for deepening the understanding of COVID-19. They also suggest the need to model the task of relation extraction at the sentence level, as opposed to the current modeling approach, developed at the abstract level. We are working to improve the system as suggested and further validate its effectiveness.

This document describes the biopatternsg's system capabilities and options of use, which we hope will facilitate the definition of adjustments and new requirements, aimed at improving the utility of the system we have called bioPatternsg (biopatterns searching). For this purpose, an example of modeling and analysis is used, in which it is desired to explore possible links between the regulatory processes inherent in SARS-COV-1, SARS-COV-2, HIV and COVID-19. The examples here only illustrate the use of the system, so their results only have an academic purpose.

This document includes two sections. The first section focuses on the interaction that the researcher typically makes with the system. The second section shows what is required to execute a modeling experiment in the context of some network, plus a trace of its execution, which describes all the resources and knowledge bases that the system organizes automatically. It will be seen that the user can with her intervention improve the results that the system provides. This document includes an annex that describes how the information used by the system is collected and organized. Please, see the wiki of the project using the link below, for updated information about how to download, install and execute the system.

Biopatternsg's repository:
https://github.com/biopattersg/biopatternsg

1. Modeling a Gene Regulatory Network. COVID-19 case

This section describes how to model a genetic regulatory network (GRN), given an initial collection of biological objects suggested for it. We run an experiment in which we want to explore possible pathways and subnets that could suggest relationships between COVID-19 and biological objects related to the immunological system. Table 1.1 shows an extract of the KB (Knowledge Base) obtained in this experiment (Please, go to the biopatternsg folder and see minery / networks/ COVID-19/ COVID-19-IMMUNOLOGY-I/kBase.pl). In addition to the KB mentioned, the system provides a documented version of it, which allows to verify the validity of the regulatory events modeled (see kBaseDoc.txt, same location). The aforementioned pair of KBs are complemented by other knowledge bases, which makes it possible to define criteria

according to which to explore the network; for example, explore objects that shape pathways that end with a particular object or disease (ACE2 or COVID-19, for instance).

1.1. On the execution of biopatternsg and the search for information

In biopatternsg, the execution of an experiment requires initial information consisting of: 1) a list of biological objects inherent to the network, in this case objects belonging to the mentioned disease, defined in the file expert_objects.txt file (see Table 1.2 and data/COVID-19/COVID-19-IMMUNOLOGY/); 2) the possible regulatory region to study; in this example the a SARS-CoV-2's RNA region (see SARS-COV-2-Regulatory file, same location); 3) an optional list of complementary objects; for example, homologous proteins (not used in this case); and 4) an optional list of PubMed IDs that the user could desire to be considered (pubmed_IDsExp file). Later we will see how this data is provided to the system. Assume that the system has already been run and that we already have the related results.

According to Figure 1, the system organizes searches by levels. For each group of objects defined at a modeling level, the system proceeds to query both the Protein Data Bank and an object identification service available through the HGNC. In the case of PDB, this service responds by informing about complexes that contain the object on which it is queried. Since PDB responds with complexes, their analysis allows obtaining new possible objects for the network; objects that become the new objects of the next level of construction of the GRN (see Figure 1). On the other hand, HGNC provides metadata about each queried object, which includes official name and related synonyms. Both queries (PDB and HGNC) offer the possibility that the list of initially provided objects is extended according to criteria defined by the user. For the time being, the option just described only allows us to obtain complexes according to the order of relevance defined by the PDB itself. Biologists and physicians are expected to propose other criteria to better exploit this functionality. Other portals are also consulted to inquire about information related to the objects of a network under construction, such as the case of GeneOntology and MeSH. More details regarding the description of the system in the paper accessible at this link. Annex A provides an introduction about this.

Table 1.1. Partial view of the regulatory events knowledge base (KB).

.

```
base([
...
event('JAK3',association,'STAT'),
event('STAT',positive_correlation,'COVID-19'),
event('CALCIUM',association,'CXCR4'),
event('TYROSINE',association,'CXCR4'),
event('ARGININE',association,'CXCR4'),
event('Aspartic Acid',association,'CXCR4'),
event('Glutamic Acid',association,'CXCR4'),
......
event('CD4',positive_correlation,'COVID-19'),
event('CD4',negative_correlation,'COVID-19'),
event('STAT',association,'CD4'),
event('JAK3',association,'CD4'),
event('CD4',positive_correlation,'STAT'),
event('CD4',association,'STAT'),
event('CXCR4',association,'CD4'),
event('CD4',association,'JAK3'),
event('CCR5',bind,'CD4'),
event('ACE2',association,'CD4'),
```

```
event('CCR5',association,'CD4'),
….
event('STAT',association,'JAK1'),
event('JAK1',association,'CXCR4'),
event('CXCR4',positive_correlation,'COVID-19'),
event('CXCR4',negative_correlation,'COVID-19'),
...
]).
```

Note: The KB was modeled automatically and contains more than 20.000 events (see kBaseG.pl in minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I) . The knowledge base kBaseG.pl is later on restricted to a more specific one, named kBase.pl, closer to the list of objects (see Table 1.2) provided at the beginning of the modeling process. Details in the sections below. The KB kBase.pl contains only 294 regulatory events (see kBase.pl in the same location).

Figure 1 shows that a KB on modeling goes through automatic analysis processes, which eventually leads to possible regulatory pathways of the style presented in Table 1.3. Table 1.3 illustrates the results in the search for pathways available in the modeled KB (see the pathways.tx in minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I for a full list). Table 1.3 shows one of several possible regulatory pathways plus the sentences that support the pathway discovered; the system provides the documented version of the KB of events, which allows locating such sentences in the abstracts processed by IA models, available through the NCBI's PubTator system. Here an example of PubTator's answer for one of the abstracts related with the pathway presented in Table 1.3.

Table 1.3 presents a pathway regarding the relationship between the objects CCR5, CD4, IFNG and ACE2 in the regulation of COVID-19. Later on we will see how to explore the set of pathways available using a graphical representation that summarizes them.

1.2. Analysis options

Table 1.4 shows the navigation view when the system is running. As Table 1.4 shows, the networks on modeling are listed first. Once we choose a network, the system lists the processes available for it. As soon as we choose any of the available processes the system lists the experiment's configuration, accompanied by a summary of the results obtained both in relation to modeling and searching for pathways. Finally, the system displays its analysis options.

Table 1.4 shows that 35 objects were analyzed for the COVID-19-IMMUNOLOGY-I experiment. We can also see the number of combinations of the objects' identifiers (names and synonyms), according to which the PubMed queries were made, which led to 9706 PubMed IDs for the experiment. Once the abstracts analysis process is made by the PubTator service, the system automatically builds the KB shown in Table 1.1. Table 1.4 reports that 20501 regulatory events were modeled for the COVID-19-IMMUNOLOGY-I experiment.

The system proposes the exploration of regulatory pathways once an experiment finishes its modeling phase. Table 1.4 shows that such a step has been executed at least once, indicating 644 inferred pathways. Once the pathways searching is complete, the main menu is available to start the network analysis phase (see Table 1.4, bottom part). The analysis menu in Table 1.4 shows, in order: 1) the possibility of executing the modeling again; 2) access the general analysis menu; 3) infer

again the regulatory pathways; 4) repeating the modeling process from a particular stage (rebuilding the KB or regulatory events, for instance). This aforementioned option makes it easier for the user to modify any of the knowledge bases and re-execute the pipeline.

Table 1.2. Initial list objects for the experiment.

JAK3
JAK1
STAT1
STAT2
STAT3
STAT4
STAT5A
STAT5B
STAT6
IPO4
IPO5
IPO7
IPO8
IPO9
IPO11
IPO13
MICA
MICB
HLA-A
HLA-B
HLA-C
CCR5
ORF6
ORF8
CD4
CD8A
CD8B
CCR5
IFNG
IFNA1
IFNB1
IFNGR1
IFNAR1
IFNA8
IFNA
IFNL1
IFN1@
IFNA3
IFN-ALPHA-14
IFNA21
CXCR4
ACE2
S Protein
COVID-19

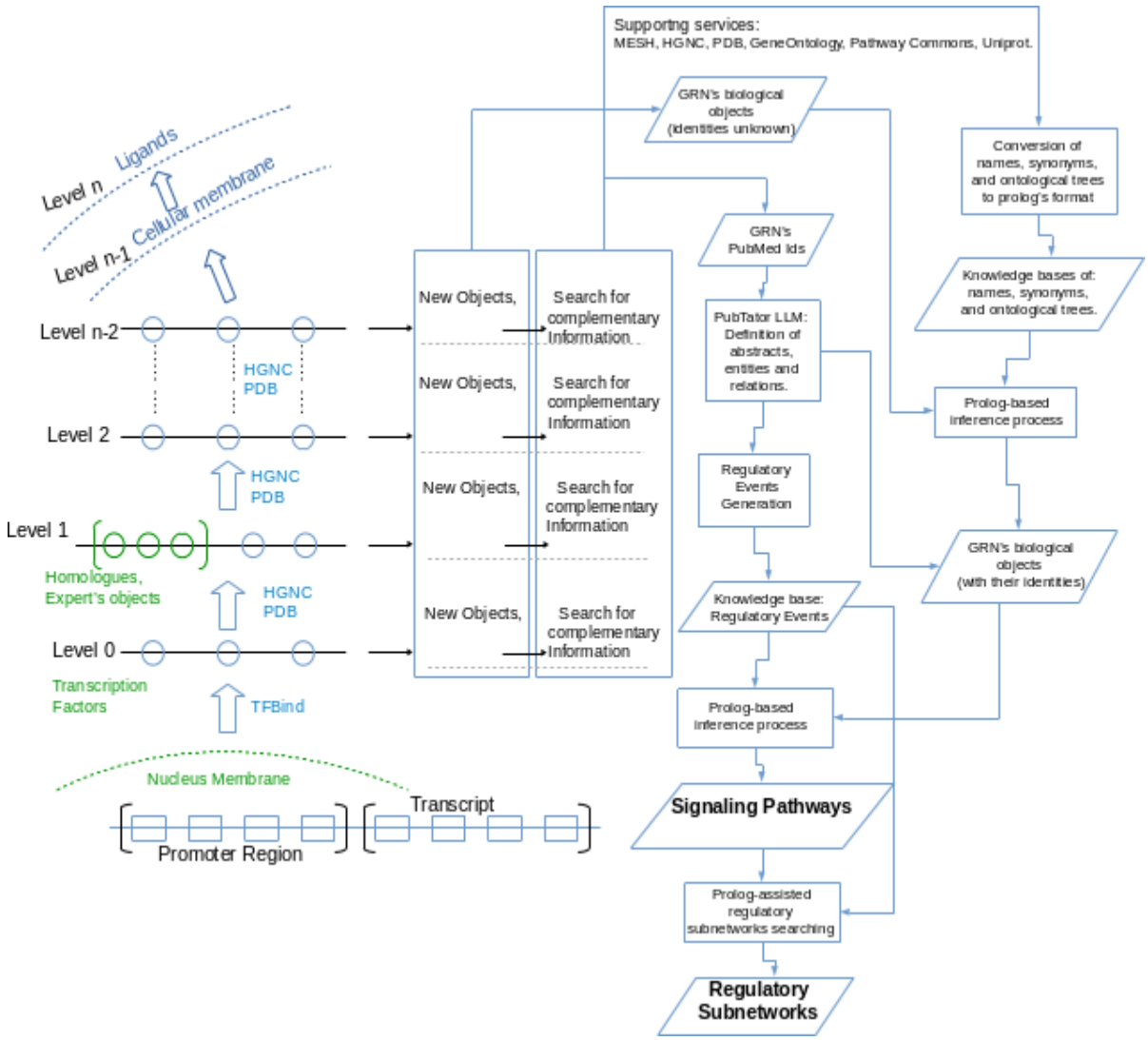

Figure 1. System architecture and work-flow for the semantic modeling and analyzing of a GRN.

Two analysis options are illustrated in this section: 1) the general search for regulatory pathways from the KB of events (see Table 1.3); and 2) the search for set of pathways (subnetworks) that first stimulate the up-regulation of an object in the network, connected to other pathways that could down-regulate it afterwards (see Table 1.5 and Table 1.6). Table 1.3 and Table 1.5 are representative of the type of outputs that the system produces, and show only part of their corresponding contents. Each of the pathways presented in such tables requires user validation of its regulatory events, using the documentation in KBaseDoc.txt. When considering the examples in this guide, it should be kept in mind that since the events' modeling is automatic, it is normal to see false positives in both pathways and binding events. At the moment, the system proposes regulatory events, pathways and subnets, according to biological restrictions that could be extended at the user's request, but this requires some additional programming in the system; those changes are not possible directly, so far.

As we mentioned before, the system provides the possibility of executing some of the modeling stages individually (see option 4, lower part of Table 1.4); Table 1.7 shows the result of accessing this option.

Table 1.3. An inferred pathway showing a correlation between the 'CXCR4' receptor and the COVID-19 disease.

```
CCR5 ---> bind
CD4 ---> association
IFNG ---> association
ACE2 ---> positive_correlation ---> COVID-19

Pathway =
    [
     event('CCR5',bind,'CD4'),
     event('CD4',association,'IFNG'),
     event('IFNG',association,'ACE2'),
     event('ACE2',positive_correlation,'COVID-19')
    ] .

CONCLUSIONS: These data support recent findings that members of the chemokine family, including CCR5 and LESTR/Fusin (CXCR4),
function as coreceptors in combination with CD4 for HIV-1 invasion.
PUBMED_ID: 9024623.

Interferon-gamma (IFNgamma) is a major component of CD4+ T cell-dependent MuHV-4 control.
PUBMED_ID: 38578092.

Among these, seven significant cDEGs and proteins - namely, HRAS, IFNG, JUN, CDH1, TLR4, ICAM1, and SCD-were recognized as pivotal
host factors linked to ACE2.
PUBMED_ID: 38640424.

In patients with COVID-19, epithelial cells showed an average three-fold increase in expression of the SARS-CoV-2 entry receptor ACE2, which
correlated with interferon signals by immune cells.
PUBMED_ID: 32591762.
```

|**Note**: The documentation of any event in a pathway can be accessed by consulting the kBaseDoc.txt file (located at minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I/. Open the kBaseDoc.txt file and search, for instance, the event event('ACE2',positive_correlation,'COVID-19').

Table 1.4 also shows the option number 2, named “Go to the GRN analysis menu”, useful for the general analysis of GRNs; Table 1.8 shows the result of accessing such an option. The option 1, visible in Table 1.8, leads to results of the type illustrated in Table 1.5 and Table 1.6, but Table 1.6 must be organized manually by the user, from the documentation available in KBaseDoc.txt. In general, the options presented in Table 1.8 provide the following query modes, so far: 1) given a pathway, find groups of linking events and inhibitory pathways related with it; and 2) given an object, display its names, synonyms, MESH and GeneOntology trees (textual, non-graphical representations, so far). Other queries are on developing right now, such as: 3) given a ligand, which receptors recognize it; 4) to identify complexes in the network and propose their possible regulatory roles; 5) given a receptor, to which complexes it binds; 6) given a ligand, to which pathways does it lead and of what regulatory type they are; 7) Given a receptor, to which DNA motifs it binds and to which pathways it leads; 8) given a receptor, which ligands recognize it; 9) given a receptor, which of its ligands could be considered agonists, antagonists or of mixed function. If you are interested in other questions please let us know.

Below we describe how to proceed to the system's execution and a detailed description of the different knowledge bases that it generates.

2. Biopatternsg. Running an experiment.

The system's products (knowledge bases) and the stages in which they are generated are briefly described here. Please do not hesitate to email us if you want our support (biopatternsg@gmail.com). If you have a way to access the system from a server, we recommend doing it now. In the case of Windows environments it is possible to use the powershell or the cmd

commands to access the server in terminal mode. We recommend the filezilla application to access the server server in explorer mode, in order to upload or download files and folders. The terminal mode is used in this guide; the explorer mode will allow you to access the different folders of the system on the server.

Table 1.4. Initial options and results when using Biopatternsg.

```
Select an option:

==== Networks available ====
1.- COVID-19

0.- Exit
1

Network: COVID-19
Select an option
==== Processes available ====
1.- COVID-19
2.- COVID-19-IMMUNOLOGY-I
3.- COVID-19-IMMUNOLOGY
0.- Go back
2

BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Initial configuration:

Regulatory region file:          COVID-19RegProm
Number of PDB objects:      1
Number of searching levels (Number of levels to build the network) :          1
Confidence for TFbind:      98
Number of PubMed IDs for every pair of combined symbols:   10
Expert's objects from:  data/COVID-19/COVID-19-IMMUNOLOGY/expert_objects.txt


Results:

Consulted objects:          35
Number of pairs of objects combined:    9706
Pubmed IDs:     3014
Regulatory events:      294
Patterns (Pathways):      2556

Select an option:
1.- New process
2.- Go to the GRN analysis menu.
3.- Go to infer pathways.
4.- Run the system from
0.- Go back
```

Biopatternsg requires some initial *data* to run an experiment in the context of a network (in this example COVID-19). The way to provide such data is to place it in a folder called ***data***, which is the folder containing the different data folders for the networks on modeling. Regarding this example, the data folder contains multiple networks, one of which is COVID-19, and that folder contains four more, named: COVID-19, COVID-19-Drugs, COVID-19-IMMUNOLOGY and COVID-19-IMMUNOLOGY-I. The data/COVID-19/COVID-19 folder refers to an experiment to model interactions of objects related to COVID-19, while data/COVID-19/COVID-19-Drugs refers to an experiment in which we want to see possible relationships between objects related to COVID-19 and specific drugs. The data/COVID-19/COVID-19-IMMUNOLOGY experiment contains names of COVID-19 objects interacting with objects related with the immune system. Generally speaking, each experiment in each data folder will contain files like these:

Table 1.5. Example of a possible stimulatory / inhibitory subnetwork for COVID-19.

Pathway=> 'CCR5',bind,'CD4';'CD4',association,'IFNG';'IFNG',association,'ACE2';'ACE2',positive_correlation,'COVID-19'
Linking events: 'COVID-19',positive_correlation,'CD4'
Pathway=> 'CD4',association,'JAK3';'JAK3',bind,'STAT3'
Linking events: 'STAT3',positive_correlation,'ACE2';
Pathway=> 'ACE2',association,'CD4';'CD4',association,'IFNG';'IFNG',association,'JAK3';'JAK3',association,'CXCR4';'CXCR4',negative_correlation,'COVID-19'

1. SARS-COV-2-Regulatory. Normally the file related to this option contains the regulatory DNA sequence of a gene and it is used by the system to explore transcription factors for it. In this example we are modelling a GRN for a disease and transcription factors are not used but normally, when you are modeling a GRN for a gene, they are.
2. expert_objects.txt. This file lists the initial set of objects for the network to model.
3. homologous (optional). This file lists the names of homologous proteins related to a protein or some proteins of interest. In this experiment the file is empty.
4. pubmed_IDsExp (optional). It corresponds to a list of PubMed IDs of abstracts that the user/expert wants to be included in the network's modeling.

2.1 An experiment, its tuning and its outputs

Table 2.1 shows how to run experiments on the system. As Table 2.1 shows, the system displays the list of networks available. Once we select a network (say COVID-19), the system lists the different modeling processes/experiments that may have been developed for it, offering the opportunity to 1) integrate them or 2) select a particular experiment. If you choose to integrate (option I), then all the available knowledge bases for the network chosen are integrated, therefore generating a single model for the entire network. In case of choosing a particular process (say 3), then the corresponding results for that experiment are displayed. In the event that no modeling experiment has been developed for the selected network, the system requests the necessary parameters to run one for it. Table 2.1 corresponds to the aforementioned case.

2.1.1 Running an experiment

If you have a local installation of the system on your Linux machine, then it is a good time to go to the folder where the installation took place. In case of accessing a server, you must have received the corresponding access details from the system's administrator.

Table 1.6. The documentation for the hypothetical regulatory subnetwork in Table 1.5 .

Pathway=> ‘CCR5',bind,'CD4';'CD4',association,'IFNG';'IFNG',association,'ACE2';'ACE2',positive_correlation,'COVID-19'

event('CCR5',bind,'CD4').
CONCLUSIONS: These data support recent findings that members of the chemokine family, including CCR5 and LESTR/Fusin (CXCR4), function as coreceptors in combination with CD4 for HIV-1 invasion.
PUBMED_ID: 9024623.

event('CD4',association,'IFNG').
Interferon-gamma (IFNgamma) is a major component of CD4+ T cell-dependent MuHV-4 control.
PUBMED_ID: 38578092.

event('IFNG',association,'ACE2')
Among these, seven significant cDEGs and proteins - namely, HRAS, IFNG, JUN, CDH1, TLR4, ICAM1, and SCD-were recognized as pivotal host factors linked to ACE2.
PUBMED_ID: 38640424.

event('ACE2',positive_correlation,'COVID-19')
In patients with COVID-19, epithelial cells showed an average three-fold increase in expression of the SARS-CoV-2 entry receptor ACE2, which correlated with interferon signals by immune cells.
PUBMED_ID: 32591762.

Linking events:
event('COVID-19',positive_correlation,'CD4').
Results: A third COVID-19 mRNA vaccine dose significantly boosted spike-specific CD4+ and CD8+ T-cell frequencies to above 2-dose levels in older and younger adults. PUBMED_ID: 38035132

Pathway=> 'CD4',association,'JAK3';'JAK3',bind,'STAT3'

event('CD4',association,'JAK3').
Of interest is the finding that the administration of the JAK3 inhibitor during acute infection also resulted in the sustained maintenance during chronic infection of a high number of naive and central memory CD4+ T cells, increases in B cells in the blood, but decreases in the frequencies and function of NKG2a+ NK cells within the GIT and blood, respectively. PUBMED_ID: 24603870.

event('JAK3',bind,'STAT3').
In contrast, Jak3 co-expression resulted in tyrosine phosphorylation of STAT3 and STAT5 but not STAT1
PUBMED_ID: 10037026,

Linking events:
event('STAT3',positive_correlation,'ACE2').
ACE2 transcription is enhanced by transcription factors including Ikaros, HNFs, GATA6, STAT3 or SIRT1, whereas ACE2 transcription is reduced by the transcription factor Brg1-FoxM1 complex or ERRalpha. PUBMED_ID: 37608279

Pathway=>
'ACE2',association,'CD4';'CD4',association,'IFNG';'IFNG',association,'JAK3';'JAK3',association,'CXCR4';'CXCR4',negative_correlation,'COVID-19'

event('ACE2',association,'CD4')
The expression of ACE2 was strongly positively correlated with the immune infiltration level of CD8+ T cell ($r=0.184$, $p<0.001$), CD4+ T cell ($r=0.104$, $p=0.02$) and neutrophils ($r=0.101$, $p=0.02$). PUBMED_ID: 34413267.

event('CD4',association,'IFNG')
Indirect CD4+ T cell protection against mouse gamma-herpesvirus infection via interferon gamma.
PUBMED_ID: 38578092.

event('IFNG',association,'JAK3')
These interacted with the IL-15 receptor complex on CD4(+) T cells, thus activating the Jak3 and STAT5 phosphorylation signalling pathway to induce CD40 ligand expression, T-cell proliferation and IFN-gamma production.
PUBMED_ID: 20391436.

event('JAK3',association,'CXCR4')
The chemokine SDF-1alpha triggers CXCR4 receptor dimerization and activates the JAK/STAT pathway. PUBMED_ID: 10506573.

event('CXCR4',negative_correlation,'COVID-19')
Interestingly, the expression of cell receptors, such as IFNGR1 and CXCR4, was reduced in response to the viral infection and associated with the inhibition of the related signaling pathways and immune functions.

Table 1.7. Options available for the re-execution of the modeling stages.

Select an option 1.- Keywords generation 2.- Search for PubMed IDs 3.- Formatting names and ontologies 4.- Generating knowledge base of events 0.- Go back

Table 1.8. Options available for the analysis of the information of a GRN.

Queries for the GRN analysis. 1.- Search for connected pathways 2.- Querying an object 0.- Go back

Once you are in the biopatternsg folder, the commands indicated below must be executed. It is recommended to run the *screen* command first, useful to retrieve the experiment in progress later on. The steps below should lead us to the view indicated in Table 2.1, which shows how to configure a new experiment; be sure to select the <COVID-19-IMMUNOLOGY> option.

The command below allows us to track the terminal in which the experiment will run.

```
$ screen
```

Change to the biopatternsg's folder:

```
$ cd ~/tools/biopatternsg
```

Then run the biopatternsg system using the following command:

```
$ ./biop.sh
```

Then, the system will show a menu to choose a language to interact with. Once you choose a language, you will see a screen similar to the one presented in Table 2.1. Below the details regarding the configuration of a modeling experiment.

- Regulatory region file name. It corresponds to the DNA sequence of the regulatory region of the main biological object in study; in this case, it is provided in the file SARS-COV-2-Regulatory file (see data / COVID-19/COVID-19-IMMUNOLOGY/SARS-COV-2-Regulatory).
- Select search source for TFs. This option allows the definition of sources to predict transcription factors able to bind to motifs in the regulatory region provided; at the moment the system provides two options: Jaspar and TFBind. In this exercise we choose option 1, meaning that only FTs from TFBind will be used.
- Confidence value for TFs search (0-100). The minimal confidence measure for the predicted TFs in order to include them in the GRN on modeling. If we select option 2 or 3 then the system consults the JASPAR service for predictions of transcription factors for the regulatory region previously provided by the user. Table 2.2 below shows parameters for the Jaspar in order to get the best candidate FTs for the regulatory region provided in the experiment. We present Table 2.2 as an instance of what normally will happen when the user is modeling a GRN and including the JASPAR service. In such a case we must provide the genome assembly, chromosome and

coordinates that better suit the regulatory DNA sequence provided in the file SARS-COV-2-Regulatory. In order to do that we use the Genome Browser's Blat service provided by the University of California Santa Cruz (genome.ucsc.edu/index.html). In this video tutorial we see how to get the parameters shown in Table 2.2).

- Number of PDB objects. The amount of protein complexes for each object on the network to download from PDB. In this case, in this experiment we ask PDB only for one complex for each object that is already part of the network; each complex is then analysed by the system and its objects are added to the network. In this example only FTs with a probability equal to or greater than 0.98 are chosen.

Table 2.1. Configuring and running an experiment

```
BioPatternsg

Select an option
==== Networks available ====
1.- COVID-19
0.- Exit
1

Network: COVID-19
Select an option
==== Processes available ====
1.- COVID-19
2.- COVID-19-IMMUNOLOGY-I
3.- COVID-19-IMMUNOLOGY
0.- Go back
3

Biopatternsg

-----------------------------
New data mining process
-----------------------------

Provide configuration data:

Regulatory region file name: SARS-COV-2-Regulatory
Select search source for TFs:
1. TFBind
2. Jaspar
3. TFBind/Jaspar
1
Confidence value for TF search (0-100): 99
Number of PDB objects (The amount of PDB objects that must be downloaded): 1
Number of searching levels (Number of levels to build the network): 1
Do you want to specify your own set of PubMed IDs? ..(Y/N): y
Set the name of the file with the PubMed IDs: pubmed_IDsExp
Number of PubMed IDs for every pair of combined symbols: 10
Only short names to combine and generate paired symbols... (Y/N):y


BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY

Searching for complementary information for the expert's objects.

search JAK.........................
search JAK3........................
```

- Number of searching levels. The number of levels that will be explored in the construction of the network model (see Figure 1). In this case, we use only one level and that means to work with the expert's objects, the transcription factors coming from JASPAR and TFBIND, and one level of extra objects coming from PDB.

- Do you want to specify your own set of PubMed IDs?. This option indicates if a list of PubMed IDs will be provided by the user and the name of the file that contains them is pubmed_IdsExp.
- Number of PubMed IDs for every pair of combined symbols. It corresponds to the number of PubMed IDs to search for each pair of keywords consulted. For instance, if the combination of JAK3 and STAT is consulted to PubMed, then Table 2.1 shows that only the first 10 more relevant PubMed IDs will be recorded and downloaded later.
- Only short names to combine and generate paired symbols. It defines whether short (only one word) or long names for the object's identifiers will be used in the searching for PubMed IDs in PubMed. It is normal for objects to have acronyms or mnemonics associated; answering yes implies that only short names (only one word) must be used.

Table 2.2. The JASPAR service asks for the best coordinates for the regulatory region .

Genome Assembly for Transcription Factors search: wuhCor1
Chromosome for Transcription Factors search: NC_045512v2
Start coordinate for Transcription Factors search: 1
Final coordinate for Transcription Factors search: 2264

Once the data is provided the experiment will start and it will take some time to finish. You should see something in the terminal like this:

```
**Searching for complementary information for the experts's objects..

search JAK.......
search JAK3......
```

The initial output of the system above regards the searching of complementary information for the initial set of biological objects listed in the file expert_objects.txt.

Once the JASPAR/TFBind service(s) provide FTs for the regulatory region, the system will go through all the stages depicted in Figure 1, until it reaches the stage in which it is ready to generate the knowledge base of regulatory events (see Table 2.3). At this point of the modeling process the user models a KB in three options: 1) General (not restricted at all, option N); 2) Restricted (option R) and 3) Very restricted (option V). The options R and V in Table 2.3 allow the modeling of a KB restricted to a list of specific objects. When the option is R each regulatory event in the KB will have at least one of the objects in the list provided, and when the option is V both of the objects in each regulatory event will be objects from the list provided. The rest of the process is the same; the only difference in the analysis process later on will be using a general or a restricted knowledge base, and if the use case in the last one, then the user has decided which objects she wants to put her attention on. Note: this exercise restricts the knowledge the option V described below and the way we proceed is as Table 2.3 describes. Finally, once the KB of regulatory events is ready, the pipeline stops waiting for instructions about how we want the inference of pathways to go on (see Table 2.4). The sections below will show more details regarding the inference of pathways and subnetworks.

The time that an experiment requires to finish will depend on the number of objects and the parameters provided; time that can vary from a few hours to some days. Note: If your experiment is stopped for any reason, the system is able to recover it from the closest stable stage to the moment of the interruption.

Table 2.3. The prompt about generating the knowledge base.

```
BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY

Generating knowledge base
Do you want to restrict the knowledge base of events (Y | N): y
Please, set the kind of restriction (R | V): V
Provide a list of objects' symbols separated by (;): for example: EGF;EGFR;Ras;CREB;SST

S;JAK3;JAK1;STAT1;STAT2;STAT3;STAT4;STAT5A;STAT5B;STAT6;IPO4;IPO5;IPO7;IPO8;IPO9;IP
O11;IPO13;MICA;MICB;HLA-A;HLA-B;HLA-C;CCR5;ORF6;ORF8;CD4;CD8A;CD8B;CCR5;IFNG;I
FNA1;IFNB1;IFNGR1;IFNAR1;IFNA8;IFNA;IFNL1;IFN1@;IFNA3;IFN-ALPHA-14;IFNA21;CXCR4;
ACE2;COVID-19
```

Table 2.4.The prompt about inferring pathways for a set of biological objects.

```
BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Pathways Inference:
Do you want to limit the objects in the pathways to a specific list?  ..(Y/N):
```

Now that the modeling process is running we can execute the following command to leave the experiment running on the server. Please hold the following combination of keys.

```
$ <Ctrl> + <a> + <d>
```

Then execute the following command to recover the experiment.

```
$ screen -r
```

Re-execute `<ctrl> + <a> + <d>` to leave the experiment running in the background.

Now it is possible to close the terminal and leave the modeling process running on the server. Please hold the combination of keys below:

```
$  <ctrl> + <d>
```

The *screen* command allows you to have multiple experiments running and retrieve them at any time. If you have more than one experiment running, then the *screen -r* command will return the identifiers associated with each one of them. If you run *screen -r <identifier>*, the command will return the terminal and the experiment related to it.

Let's now run the system again, but now to explore an example of what you will get at the end of the experiment that you just started.

```
$ screen
$ cd ~/tools/biopatternsg
$ ./biop.sh
```

Navigate the menus and select the option <COVID-19-IMMUNOLOGY-I> from the output illustrated in Table 2.1.

At this point the system's output in your terminal should look like Table 2.4, showing the stage in which it is possible to infer pathways.

2.1.2 Adjusting the knowledge bases to improve an experiment's results

Before we get results like those in Table 1.3 and 1.5, some adjustments must be made to the knowledge that the system organizes automatically. Here we illustrate such changes using the knowledge base that describes the roles of the objects in a GRN model and these roles guide the search for pathways. The file in this example is named pathwaysObjects.pl, see an excerpt of it in Table 2.5.

Regarding the pathwaysObjects.pl file, the user must verify if the roles automatically assigned to the objects in the network are correct. Roles such as ligand, protein, transcription factor or receptor can be seen in the aforementioned file (see the pathwaysObjects.pl in the experiment's folder). The user can play with the roles, thus defining which objects can start pathways and which can close them. Normally the user will have to modify the pathwaysObjects.pl KB, in order to get better results in an experiment. Table 2.5 is an excerpt of a file that has been prepared as a reference; the file is saved in the COVID-19-IMMUNOLOGY-I experiment's folder and is named pathwaysObjects.bkp. In order to get the same results illustrated in Table 1.3 and Table 1.5, please open up the file pathwaysObjects.bkp and save it as pathwaysObjects.pl.

Now the user is able to explore the inference of pathways and subnets.

2.1.3 Inferring pathways and subnets

Table 2.6 shows that a list of biological objects and drugs is provided to the system. Then, we specify to the system some objects that we want to see as the objects that are up-regulated or down-regulated in the pathways inferred. First let's

say we want to reproduce the results shown in Tables 1.3 and 1.5, using the KBs in the COVID-19-IMMUNOLOGY-I experiment. The procedure would be as follows:

Table 2.5. The knowledge base that describes the role of the objects in a GRN model

```
disease('COVID-19').
disease('Severe Acute Respiratory Syndrome').
transcription_factor('STAT1').
protein('STAT1').
transcription_factor('STAT').
protein('STAT').
transcription_factor('STAT3').
protein('STAT3').
...
..
transcription_factor('STAT2').
protein('STAT2').
receptor('CXCR4').
protein('CXCR4').
protein('ACE2').
enzyme('ACE2').
receptor('ACE2').
receptor('CD4').
protein('CD4').
ligand('CD4').
...
```

To reproduce the results illustrated in Table 1.3, we must provide to the system the parameters indicated in Table 2.6. Then, the inference of pathways must start. Once the inference of pathways is finished, the system will display a summary of the experiment's parameters and the number of pathways found (in this case 2556). At this point your terminal should look at the bottom like Table 2.6.

The inference of pathways generates two main files 1) pathways.txt, which contains the 644 pathways found; and 2) kb_pathways.pl, which contains the events that shape the different pathways inferred.

Table 1.3 corresponds to one of the 2556 pathways listed in pathways.txt. In order to find the pathway described in Table 1.3 in the file pathways.txt, we recommend opening that file and following the note at the bottom of Table 2.6. Once we have the pathway, if we want to get the sentences and the related PubMed IDs, we must copy each event, open the file kBaseDoc.txt, and search for them. Let's say we do that for the event 'ACE2',positive_correlation,'COVID-19', the result will be as Table 2.7 shows. Table 2.7 shows the regulatory event, the different ways the objects ACE2 and COVID-19 are named in the abstracts from PubTator, and the set of sentences that supports the regulatory event 'ACE2',positive_correlation,'COVID-19'; normally there are a set of sentences for each regulatory event but, in this case, there is only one.

Now let's reproduce the regulatory subnetwork in Table 1.5.

First we must choose option 2 from the lower menu in Table 2.6, to get access to the general menu regarding the analysis of a network and that must take us to the menu shown in Table 1.8. Once we are there, select option 8, related to “search of connected pathways”; now Table 2.8 should be the new menu to follow.

Table 2.6. Inferring pathways for a set of biological objects.

```
BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Pathways Inference:
Do you want to limit the objects in the pathways to a specific list?  ..(Y/N):n
Do you want to provide a symbol object to finish the pathways? ..(Y/N): y
Provide the object name: STAT3
Do you want to provide another object to finish the pathways?.. (Y/N): y
Provide the object name: ACE2
Do you want to provide another object to finish the pathways?.. (Y/N): y
Provide the object name: COVID-19
Do you want to provide another object to finish the pathways?.. (Y/N): n
Do you want to use a reduced knowledge base?  .. (Y/N):n

BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Pathways Inference
...............................................
...............................................


BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Initial configuration:

Regulatory region file:              COVID-19RegProm
Number of PDB objects:          1
Number of searching levels (Number of levels to build the network) :              1
Confidence for Transcription Factors search:           98
Number of PubMed IDs for every pair of combined symbols:   10
Expert's objects from:  data/COVID-19/COVID-19-IMMUNOLOGY/expert_objects.txt

Results:

Consulted objects:            35
Number of pairs of objects combined:    9706
Pubmed IDs:       3014
Regulatory events:        294
Patterns (Pathways):       2556


Select an option
1.- New process
2.- Go to the GRN analysis menu.
3.- Go to infer pathways.
4.- Run the system from
0.- Go back
```

Note: Search the pathway described in Table 1.3 in the file pathways.txt, using the next array of names (copy the array as it is):

['CCR5', 'CD4', 'IFNG', 'ACE2', 'COVID-19']

Table 2.7. The documentation for the regulatory event 'ACE2',positive_correlation,'COVID-19' from kBaseDoc.txt.

********** Regulatory Event **********

event('ACE2',positive_correlation,'COVID-19')

subject names: ['ACE2', 'Angiotensin-Converting Enzyme 2', ... 'Angiotensin converting enzyme 2']
object names: ['COVID-19', 'severe coronavirus disease 2019', 'severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection', ..., 'severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infections']

Sentences from abstracts:

In patients with COVID-19, epithelial cells showed an average three-fold increase in expression of the SARS-CoV-2 entry receptor ACE2, which correlated with interferon signals by immune cells. PUBMED_ID: 32591762

As Table 2.8 shows, we copy from Table 1.5 the first pathway shown there and supply it as the starting point to infer subnetworks. This pathway was originally chosen from the pathways.txt file, suggesting that the user found something of interest in it; which motivates her to use it as a starting pathway when inferring regulatory subnetworks. Once we provide the initial pathway, the system will ask about how many objects you will allow to be present in the pathways in the subnetworks. It is usually a number between 3 to 8, but it can be any number; it is recommended to gradually increase the number until you get some interesting result (we used 5). Once the inferring of regulatory subnetworks has finished, all the subnetworks found are saved in a file called chainsPathways.txt. We recommend to rename the file chainsPathways.txt so that you do not lose it in future executions of this system's option; here we have chosen to rename the file using the object that closes the pathways; in this case COVID-19 (see the file COVID-19_chainsPathways.txt). Note that the subnetworks are sets of pathways that apparently first stimulate the presence of the COVID-19 disease and then its inhibition. If you open up the file COVID-19-chainsPathways.txt, you will see many different regulatory subnetworks; this happens as there are a lot of different ways to link the pathways using linking regulatory events. The way to reduce such amount on subnetworks is to use restrictions of two main types: 1) restrictions about the objects that will shape the pathways in pathways.txt and 2) restrictions about the properties of the pathways and linking events that will shape an interesting subnetwork to search for in the file COVID-19-chainsPathways.txt. See López et al. 2020, for examples regarding the use of restrictions in biopatternsg. Once you have found an interesting regulatory subnetwork it is possible to retrieve the documentation that supports it; just proceed in the same way we did for the pathway shown in Table 1.3, that process will lead you to your own version of Table 1.6.

Now, let's go back to the experiment you started earlier. To do this, exit the current experiment by going back in the menus. And close the terminal in which the current experiment is running, using the command <ctrl> + <d>. Now retrieve the experiment for <COVID-19-IMMUNOLOGY> with screen -r. If you disconnect yourself from the server by mistake, reconnect with the server again. Your experiment will not be lost.

Table 2.8. The menu to configure the searching for regulatory subnetworks

```
BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Searching for pathways and regulatory events that relate them:

Select an option
1.- Run the process
0.- Go back
1

Provide the start pathway:'CCR5',bind,'CD4';'CD4',association,'IFNG';'IFNG',association,'ACE2';'ACE2',positive_correlation,'COVID-19'
Provide the maximum number of objects allowed in the pathways: 8
***'CCR5',bind,'CD4';'CD4',association,'IFNG';'IFNG',association,'ACE2';'ACE2',positive_correlation,'COVID-19'

Subnetworks:
------------------------------------------------------------
Pathway=> 'CCR5',bind,'CD4';'CD4',association,'IFNG';'IFNG',association,'ACE2';'ACE2',positive_correlation,'COVID-19'

Linking events:  'COVID-19',association,'CD4'; 'COVID-19',positive_correlation,'CD4'; 'COVID-19',negative_correlation,'CD4';

Pathway=> 'CD4',association,'JAK3';'JAK3',bind,'STAT3'

Linking events:  'STAT3',negative_correlation,'JAK3';

Pathway=> 'JAK3',association,'STAT6';'STAT6',association,'CXCR4';'CXCR4',negative_correlation,'COVID-19'

------------------------------------------------------------
Pathway=> 'CCR5',bind,'CD4';'CD4',association,'IFNG';'IFNG',association,'ACE2';'ACE2',positive_correlation,'COVID-19'

Linking events:  'COVID-19',association,'CD4'; 'COVID-19',positive_correlation,'CD4'; 'COVID-19',negative_correlation,'CD4';

Pathway=> 'CD4',association,'JAK3';'JAK3',bind,'STAT3'

Linking events:  'STAT3',negative_correlation,'JAK3';

Pathway=> 'JAK3',association,'STAT6';'STAT6',association,'CXCR4';'CXCR4',association,'CD4';'CD4',negative_correlation,'ACE2'

Linking events:  'ACE2',positive_correlation,'MICA';

Pathway=> 'MICA',association,'HLA-B';'HLA-B',association,'CD4';'CD4',negative_correlation,'COVID-19'

------------------------------------------------------------


......
......
```

Your experiment may not be over yet, so you must be in one of the stages corresponding to the process of gathering information. Below we describe a typical trace that will help you to locate the stage in which your experiment is and the type of product that it generates. If when entering the experiment you find it interrupted, rerun the experiment; the system will resume it at the closest stage to the point of the failure. In what follows, all the stages are described and supported by their corresponding product, just as if you were observing your experiment after it has finished.

2.1.4 Products generated in the exploration of knowledge

Table 2.1 shows that the information searching process starts when we set the parameters for an experiment; Table 2.9 shows a simplified trace of what happens afterwards using the COVID-19-IMMUNOLOGY-I experiment. Table 2.9

illustrates the different modeling stages as they unfold. The data in Table 2.9 is commented below remarking the products (knowledge bases and files) of each stage. The system products for this example are available in the biopatternsg's folder (see minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I). Below a description of the trace shown in Table 2.9.

- (optional) Consulting homologous objects. It corresponds to the search for complementary information for the homologues associated with the main object for the experiment (in this case SARS-COV-2). This option is not used in this example because it has no biological sense, so Table 2.9 does not show any information about it.

- Consulting expert objects information. It corresponds to the search for complementary information for the objects that the expert provides in the file experts_objects.txt. The search for complementary information includes the official names of the objects in the network, their synonyms, the tissues in which they are present, uniprot codes, MESH and GeneOntology ontologies for each network's objects, among other details.

- Consulting FTs from JASPAR/TFBind. It corresponds to the search for complementary information, referring to each of the FTs predicted using JASPAR and TFBind. Note: The complementary information consulted for any object on the network is saved in databases (files with extension .db) and files like minedObjects.txt. minedObjects.pl, ontologyMESH.pl and ontologyGO.pl, well_known_rules.pl and others.

- Generating keyword combinations. Once the network objects are organized, all the names and synonyms found are used to generate the collection of keywords used later on for the searching of PubMed Ids.

- Search for PubMed IDs. In this stage the keyword combinations generated before are used to define and to record PubMed IDs of abstracts, related with the objects of the network and their interactions.

- Getting annotated abstracts from PubTator. Once we have the PubMed IDs, they are used to consult the NCBI`s PubTator service, in order to collect annotated abstracts with data about the biological entities in them and their corresponding relationships. The annotated abstracts are saved in the experiment's folder named abstracts. Several files with .txt extension are available in the abstracts folder with the names abstracts, objects and relations; each of them keeps the abstracts' text, the biological entities and their relationships; they are saved in that way for programmatic purposes.

- Formatting ontologies, names and synonyms. In this stage all the information collected for all the network's objects is represented in prolog's format. This stage generates the files: minedObjects.pl, pathwaysObjects.pl, ontologiaGO.pl, ontologiaMESH.pl, and well_know_rules.pl. The aforementioned files describe, for each object: 1) names and synonyms, tissues, unitpro code, among other details; 2) identity information for the object (receptor, enzyme, FT, ligand, etc.); 3) the gene ontology trees of molecular function, biological processes and cellular components; 4) the object's identity definition and families, according to MESH; and, 5) the representation of identity information from MeSH in the form of rules A if B, to support automatic analysis using prolog.

Table 2.9. Trace of the system's execution (COVID-19-IMMUNOLOGY-I experiment).

```
BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

**Searching for complementary information for the experts's objects..

search JAK.........................
search JAK3.........................

* Predicting FTs from JASPAR/TFBIND....
* Searching for complementary information for the FTs predicted..
* Generating keywords combinations..
* Getting PubMed IDs ..
* Getting annotated abstracts from PubTator ..
* Formatting ontologies and objects' names to prolog's format..
*Generating knowledge base
  Do you want to restrict the knowledge base of events (Y | N): y
  Please, set the kind of restriction (R | V): V
  Provide a list of objects' symbols separated by (;): for example: EGF;EGFR;Ras;CREB;SST

S;JAK3;JAK1;STAT1;STAT2;STAT3;STAT4;STAT5A;STAT5B;STAT6;IPO4;IPO5;IPO7;IPO8;IPO9;IP
O11;IPO13;MICA;MICB;HLA-A;HLA-B;HLA-C;CCR5;ORF6;ORF8;CD4;CD8A;CD8B;CCR5;IFNG;I
FNA1;IFNB1;IFNGR1;IFNAR1;IFNA8;IFNA;IFNL1;IFN1@;IFNA3;IFN-ALPHA-14;IFNA21;CXCR4
;ACE2;COVID-19

Pathways Inference:
Do you want to limit the objects in the pathways to a specific list?  ..(Y/N): n
Do you want to provide a symbol object to finish the pathways? ..(Y/N): y
Provide the object name: STAT3
Do you want to provide another object to finish the pathways?.. (Y/N): y
Provide the object name: ACE2
Do you want to provide another object to finish the pathways?.. (Y/N): y
Provide the object name: COVID-19
Do you want to provide another object to finish the pathways?.. (Y/N): n
Do you want to use a reduced knowledge base?  .. (Y/N): n


Pathways Inference
.......
.......

BioPatternsg
minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I

Initial configuration:

Regulatory region file:              COVID-19RegProm
Number of PDB objects:          1
Number of searching levels (Number of levels to build the network) :              1
Confidence for  Transcription Factors search:           98
Number of PubMed IDs for every pair of combined symbols:   10
Expert's objects from:  data/COVID-19/COVID-19-IMMUNOLOGY-I/expert_objects.txt

Results:

Consulted objects:            35
Number of pairs of objects combined:    9706
Pubmed IDs:       3014
Regulatory events:        294
Patterns (Pathways):       2556

Select an option
1.- New process
2.- Go to the GRN analysis menu.
3.- Go to infer pathways.
4.- Run the system from
0.- Go back
```

- Generating knowledge bases. In this stage the knowledge base of regulatory events is created, from the annotated files available in the abstracts folder (see minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I/kBase.pl). In this stage the system also generates the documented version of knowledge base kBase.pl. (see kBaseDoc.txt), which makes it possible to get the sentences in the abstracts coming from PubTator, that explains each regulatory event. There is another file available from this stage named biotypes.pl, that summarizes all the biological entities present in the KB kBase.pl; that file lets the user know the identity of the biological entities in kBase.pl, no from MeSH, but from the PubTator service. The biotypes.pl file also allows the user to know the ligands (small molecules) present in the knowledge base of regulatory events.
- Pathways inference. This stage allows (firstable) to restrict the search of pathways to a particular set of the objects (optional). This option also allows the definition (secondly) of particular closure objects for the pathways to predict. It is also possible to indicate the use of a reduced version of the kBase.pl; reduced in the sense that synonym events are removed from the KB, which reduces the searching space. If the user makes the decision for a reduced knowledge base, then the new KB of regulatory events named kBaseR.pl is generated. The files generated in this stage are pathways.txt and kb_pathways.pl. The file pathways.txt includes all the regulatory pathways inferred in this stage, and the file kb_pathways.pl regards all the regulatory events that shape the regulatory pathways listed in pathways.txt. The file kb_pathways.pl is useful when we want to get a graphical view of the pathways inferred. See how to get the graphical view of the pathways below.
- Once the inference of pathways has finished a summary about the modeling and inference processes is reported, and the general menu of the system is displayed (see Table 2.9, bottom side).

2.1.5 How to draw and use a graphical representation for a set of inferred pathways

The file kb_pathways.pl becomes available once we get a set of inferred pathways and it makes it possible to produce a graphical representation, useful to identify regulatory pathways of particular interest. Below the procedure to generate a graph for the set of pathways in the file pathways.txt, using the knowledge base stored in kb_pathways.pl (see Figure 2):

1. Navigate to the prolog code named eventGraph.pl that will produce the graph following this link.
2. Open up the file kb_pathways.pl and make a copy of the knowledge base in it, then paste it in eventGraph.pl replacing the KB already available in it (Figure 2, section A).
3. From the bottom side of the code in eventGraph.pl, proceed to copy the command to produce the graph. Copy the command:

   draw_graph(1, 100, X), Y = circo(X).

4. Paste the command and then click <RUN!> (Figure 2, section B).
5. Save the graph in a place of your preference, following the options available as described in Figure 2, section C.
6. Open up the new graph with an app of your preference (your graph should be quite similar to the one in Figure 3). You can see the graph as KB-PATHWAYS-WIDE.png in minery/networks/COVID-19/COVID-19-IMMUNOLOGY-I. See alternative view in KB-PATHWAYS-CIRCLE.png, same location.
7. Analyse the graph and search for a pathway of your interest.
8. Once you have a pathway proceed to find its documentation, as described in previous sections of this guide.

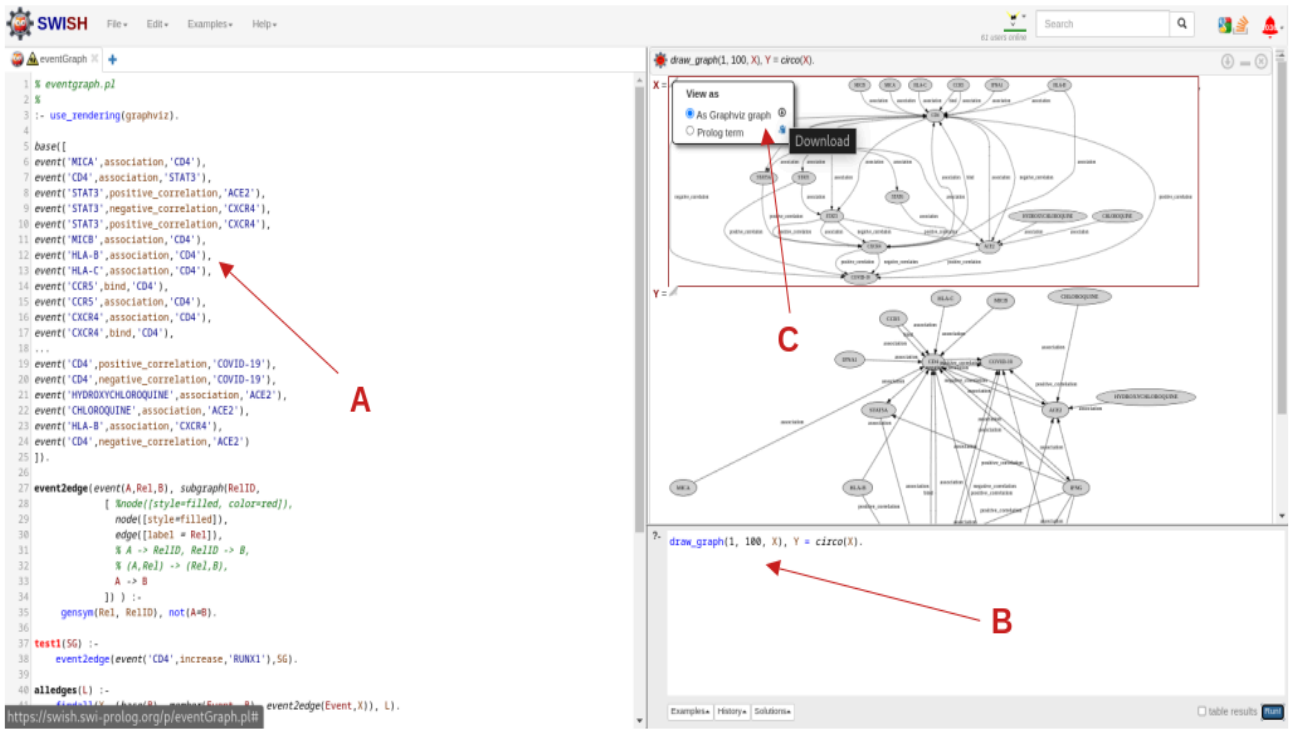

Figure 2. Drawing and saving the graph to visualize the pathways inferred

2.2 The role of the ontologies in Biopatternsg and how to take advantage of them

In Biopatternsg, we follow a knowledge representation strategy that seeks to facilitate its automatic and semi-automatic analysis; and we hope this will facilitate the incorporation of new restrictions (ways of navigating the knowledge), according to different kinds of requirements. If a user proposes a new restriction, and therefore, a different way to navigate the knowledge bases, then such a way should be relatively easy to incorporate into the system options (we hope so). About this, consider the following example.

Figure 4 shows the result of a MESH query regarding the Somatostatin hormone (SST). There you can see the six taxonomic branches, which according to MESH, correspond to the aforementioned protein. BioPatternsg uses internal definitions to represent what is described in Figure 4, in the following style: is_a ('Somatostatin-28', 'Somatostatin'), is_a ('Somatostatin', 'Pancreatic Hormones') and is_a ('Pancreatic Hormones', 'Hormones'). Such definitions (plus others of that style) allow the system to deduce that somatostatin is a family of pancreatic hormones, of which somatostatin-28 turns out to be one of its members. This approach makes it easy for the system to automatically determine if somatostatin is a hormone or not; which can lead in a searching process to a solution in which such protein participates. Note that in order to deduce that somatostatin belongs to the protein family, one must observe the different taxonomic branches that MESH

provides. In BioPatternsg, such a task is performed using automatic inference processes using knowledge bases like the one in Table 2.9.

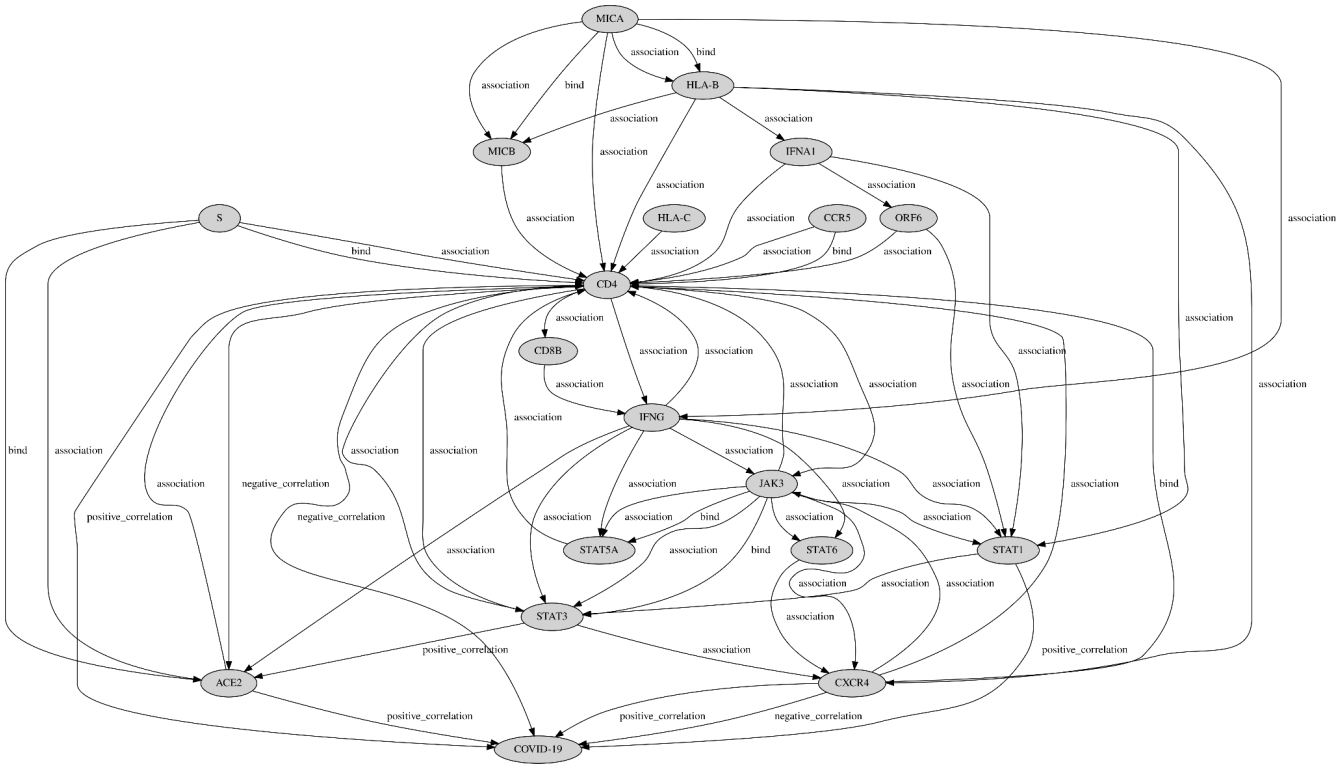

Figure 3. Graph for the knowledge base in kb_pathaways.pl (useful for the analysis of the pathways inferred).

Table 2.10 Logical representation for the Somatostatin-28 identity tree from MeSH (automatically generated)

OBJ

```
is_a('SST','Somatostatin-28').
is_a('Somatostatin-28','Somatostatin').
is_a('Somatostatin','Pituitary Hormone Release Inhibiting Hormones').
is_a('Pituitary Hormone Release Inhibiting Hormones','Hypothalamic Hormones').
is_a('Hypothalamic Hormones','Peptide Hormones').
is_a('Peptide Hormones','Hormones').
is_a('Hormones','Hormones, Hormone Substitutes, and Hormone Antagonists').
is_a('Hormones, Hormone Substitutes, and Hormone Antagonists','Chemicals and Drugs Category').
is_a('Hormones, Hormone Substitutes, and Hormone Antagonists','Physiological Effects of Drugs').
is_a('Physiological Effects of Drugs','Pharmacologic Actions').
is_a('Pharmacologic Actions','Chemical Actions and Uses').
is_a('Chemical Actions and Uses','Chemicals and Drugs Category').
is_a('Peptide Hormones','Peptides').
is_a('Peptides','Amino Acids, Peptides, and Proteins').
is_a('Amino Acids, Peptides, and Proteins','Chemicals and Drugs Category').
is_a('Hypothalamic Hormones','Neuropeptides').
is_a('Neuropeptides','Peptides').
is_a('Neuropeptides','Nerve Tissue Proteins').
is_a('Nerve Tissue Proteins','Proteins').
is_a('Proteins','Amino Acids, Peptides, and Proteins').
is_a('Somatostatin','Pancreatic Hormones').
is_a('Pancreatic Hormones','Peptide Hormones').
```

Table 2.10 shows how BioPatternsg internally represents the taxonomies received from MESH, and that is what we name the identity tree of a protein. Figure 5 shows an example, elaborated manually, that corresponds to the graphic

representation of the taxonomic knowledge received from MeSH, as a set of relationships of the form “is-a”. Please note that Figure 4, Figure 5 and Table 2.10 are equivalent regarding the knowledge that they represent. Since the system makes those trees available to all objects in a network, then the user might see opportunities from them, to design ways to restrict the kind of objects that will shape the GRN she is modeling.

The knowledge trees described above are also available for GeneOntology. Figure 6 shows part of the GeneOntology tree for somatostatin. Figure 6 only shows six of the possible twenty immediate nodes to somatostatin at the bottom; most of them associated with biological processes. We invite the reader to imagine the breadth and complexity of the corresponding graph. In bioPatternsg, trees are available logically, not graphically; however, all the potential that these graphs suggest is within the user's reach to define search criteria (or restrictions). Consider all trees for all objects in a network. If criteria of interest are defined to explore this “forest”, then possible interrelationships between objects in the network could be discovered, leading to possible solutions; for example, very specific types of pathways or subnets, restricted to very specific types of objects. The working dynamic that we propose here is one in which the user knows the potentialities of the system explained so far and, based on them, she could define new requirements and restrictions, in order to guide explorations supported by the knowledge trees described above.

All MeSH Categories
Chemicals and Drugs Category
Hormones, Hormone Substitutes, and Hormone Antagonists
Hormones
Peptide Hormones
Hypothalamic Hormones
Pituitary Hormone Release Inhibiting Hormones
Somatostatin
Somatostatin-28

All MeSH Categories
Chemicals and Drugs Category
Hormones, Hormone Substitutes, and Hormone Antagonists
Hormones
Peptide Hormones
Pancreatic Hormones
Somatostatin
Somatostatin-28

All MeSH Categories
Chemicals and Drugs Category
Amino Acids, Peptides, and Proteins
Peptides
Neuropeptides
Hypothalamic Hormones
Pituitary Hormone Release Inhibiting Hormones
Somatostatin
Somatostatin-28

All MeSH Categories
Chemicals and Drugs Category
Amino Acids, Peptides, and Proteins
Peptides
Peptide Hormones
Hypothalamic Hormones
Pituitary Hormone Release Inhibiting Hormones
Somatostatin
Somatostatin-28

All MeSH Categories
Chemicals and Drugs Category
Amino Acids, Peptides, and Proteins
Peptides
Peptide Hormones
Pancreatic Hormones
Somatostatin
Somatostatin-28

All MeSH Categories
Chemicals and Drugs Category
Amino Acids, Peptides, and Proteins
Proteins
Nerve Tissue Proteins
Neuropeptides
Hypothalamic Hormones
Pituitary Hormone Release Inhibiting Hormones
Somatostatin
Somatostatin-28

Figure 4. MESH response to a consultation inherent to Somatostatin.

2.3 How to scale the experiment described so far

Table 1.2 indicates some objects to shape a GRN model for COVID-19. On the other hand, Tabla 2.5 lets us see that different parameters values define different experiments that shape modeling processes with different orientations. Consider a scenario in which we choose to run three additional modeling processes for COVID-19. The first process could be focused on an expanded list of the objects. In the second experiment, we could keep the list shown in Table 1.2, but choose to decrease the JASPAR/TFBind reliability index about FT's predictions. The third modeling process could focus on a detailed list of drugs the user considers may inhibit the viral load of COVID-19. Once the modeling processes above complete, they could be integrated to use all the knowledge they have. Each experiment demands computing resources, so it is convenient to design different experiments that are integrated afterwards. Also consider that when we analyze results, one can carry out the experiments separately and, after having "matured" the results obtained in each case, we can proceed to use the possible advantages from the integration.

We have exposed the functionalities of a system that allows the organization of knowledge bases, starting from a set of identifiers of interest. Once the knowledge is modeled, it becomes possible to access a set of options that explore possible regulatory pathways and subnets, among other possibilities. The dynamics described here is one in which the systematization of processes is sought, but that leaves different possibilities open, so that the user actively participates. Our intention with this document has been to expose to potential users the facilities of BioPatternsg, in order to invite them to team up with us and explore possible working scenarios.

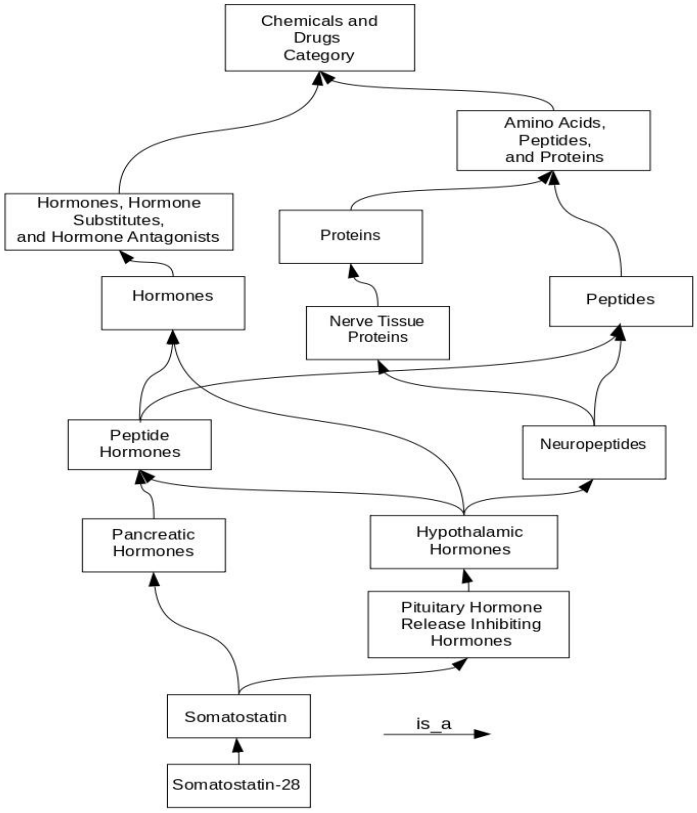

Figure 5. MESH response for Somatostatin in Biopatternsg (manually elaborated from Table 2.10).

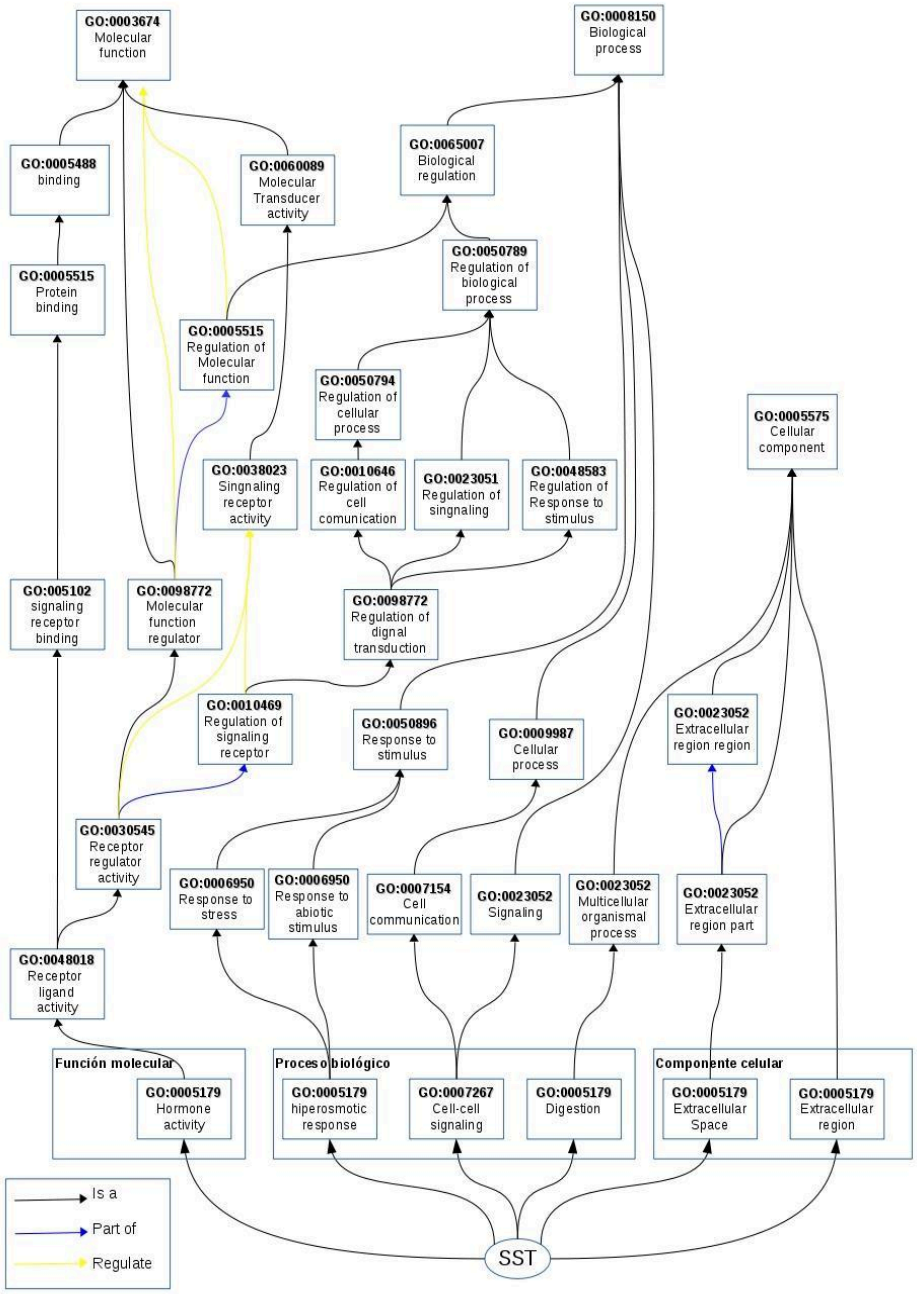

Figure 6. Partial view of the GeneOntology tree for somatostatin from biopatternsg (elaborated manually).

Annex A. Biopatternsg. About how the information is gathered

Gene Regulatory Networks (GRNs) define a very active area in the computer modeling of biological systems, and are suggesting innovative solutions to very diverse problems (Emmert-Streib et al, 2014). In such networks, a particular product (a transcript) is related to the regulation of other objects present in the same network or in different networks. Therefore, a regulatory event can activate a product that in turn participates in an event, which activates or inhibits another product. It is normal, then, that in such networks the complexity of the interrelationships grows very rapidly when we try to model them. To manage the current knowledge inherent in GRNs, computer strategies have been developed for the description, organization, interrelation and analysis of the elements that constitute them. Among such strategies are ontologies (Demir E. et al, 2010; Aditya and Babita, 2017; Muñoz-Torres and Carbon, 2017) and process diagrams (Kitano et al, 2005; Kitano,2015; Kitano, 2016); the first oriented to the semantic analysis of networks and the second to the simulation of their molecular dynamics. It is of our particular interest here, modeling and analysis systems based on inferential knowledge base processing (Rougny et al, 2018). Elaborating on such trends we have proposed the automatic construction of knowledge bases (KBs), based on the way they are described in the scientific literature. Specifically, we have developed a system that reads scientific summaries of documents and produces representations of regulatory events, for a GRN of interest, that can be computationally analyzed later. To do this, we have proposed to automatically collect the information that may be required about the objects present in a network and their possible modes of interaction.

We aim in this work: 1) the automatic construction of knowledge bases that describe GRNs (molecular species and their possible interactions) and 2) the development of exploratory analysis strategies that could guide new findings, or new conclusions, from what has already been published. Regarding 1) the developed system collects information from various repositories and services available on the Internet, necessary to answer various queries, of which we have implemented some prototypes. Regarding 2), we have designed a set of representations for the collected information, which we hope will facilitate its automatic and semi-automatic analysis. Our purpose is to answer queries of the following style: given a ligand, what kinds of protein-protein interactions result when that ligand binds to a known receptor, and which lead to activation (or inhibition) of the transcriptional response of some gene?. We also consider questions such as: for a given pair of proteins, is there a subnet in this GRN that describes interconnected regulatory pathways, in which such proteins stimulate and inhibit their transcription in any way?.

For the purposes described above, we have developed the schematic strategy expressed in Figure A.1. The modeling of a GRN starts from the region of transcription regulation of some biological object, advancing by levels, until reaching extracellular objects (a drug, for example). Our methodology begins by proposing transcription factors that could recognize the regulatory region of a protein, or other biological object (a virus, for example). From there, the modeling moves upwards integrating other objects, provided by the user or consulted on the Internet. Figure 1 shows that other objects can be added to the network from services such as those provided by the Protein Data Bank (PDB) (Rose et al, 2017) (Berman et al, 2014).

Our searching for regulatory pathways is guided by definitions like this: a regulatory pathway is one in which a ligand recognizes a receptor, which triggers a cascade of regulatory events in which different types of proteins may be involved;

cascade of events that is closed by one in which a transcription factor recognizes a response element, and therefore activates or inhibits the transcription of a specific product (RNA or protein). In our case, the finding of the solution consists in determining a collection of regulatory events that satisfy the restrictions implicit in the previous definition. Having organized knowledge automatically and semi-automatically, as indicated in Figures A.1 and A.2, then events that satisfy specific constraints can be explored in a knowledge base like the one depicted in Table A.1. Our goal is to develop a logical and ontological framework, in which it may be possible to receive definitions from the user (like the previous one), program them and build knowledge bases, robust enough to help in their resolution.

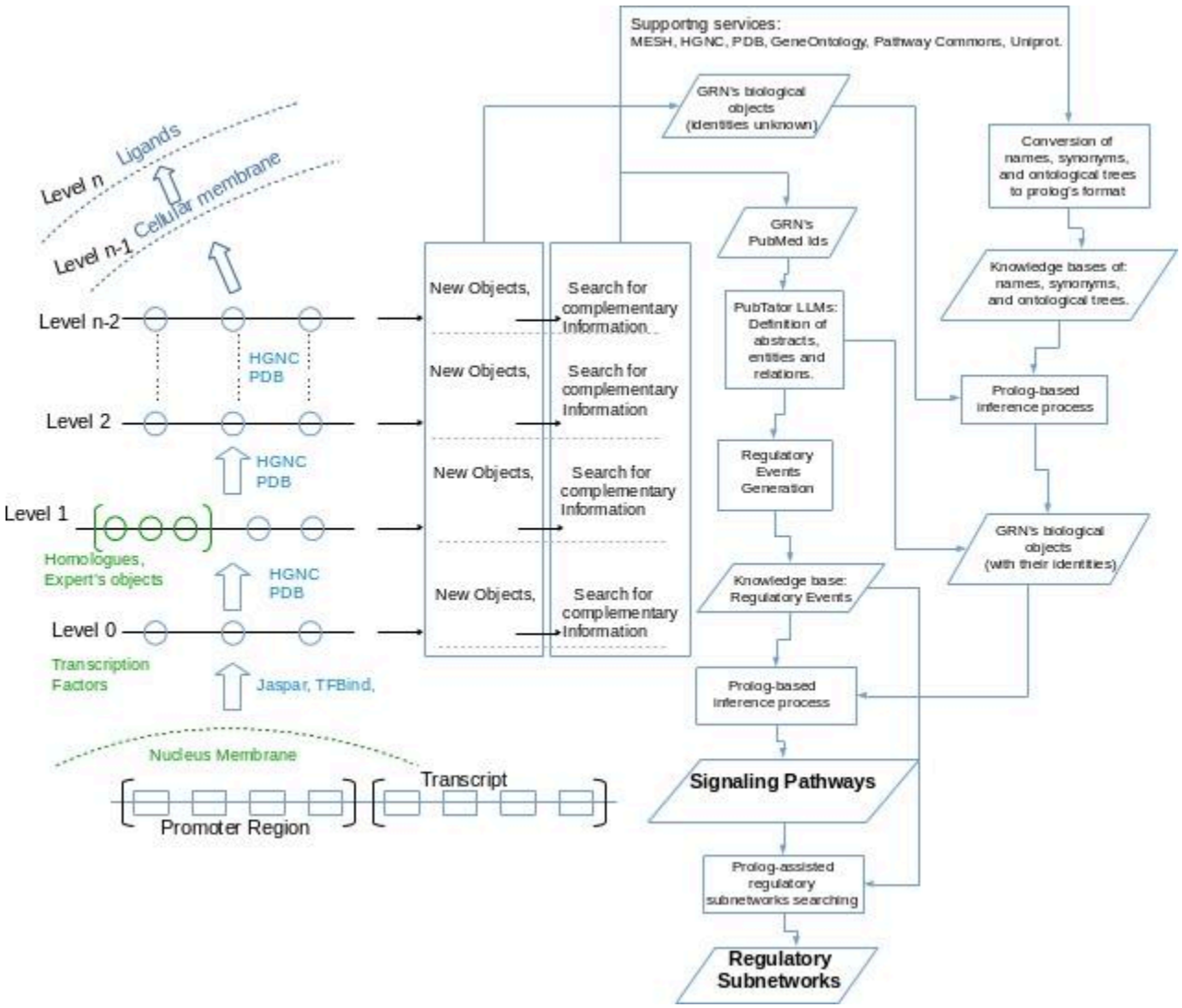

Figure A.1. System architecture and work-flow for the semantic modeling and analyzing of a GRN.

Table A.1. A subset of regulatory events in the BAXS System GRN.

```
base([
event('cholesterol',regulate,'oxysterols'),
event('oxysterols',bind,'LXRa'),
event('oxysterols',activate,'LXRa'),
event('LXRa',associate,'RXR'),
event('LXRa',bind,'LXRE'),
event('LXRa',activate,'CYP7A1'),
event('CYP7A1',increase,'ba'),
event('ba',bind,'FXR'),
event('ba',activate,'FXR'),
event('FXR',associate,'RXR'),
event('FXR',bind,'FXRE'),
event('FXR',regulate,'SHP'),
event('SHP',associate,'LRH'),
event('LRH',bind,'LRHRE'),
event('LRH',inhibit,'SHP'),
event('LRH',inhibit,'CYP7A1')
]).
```

Nota: This KB has been modeled from the literature and corresponds to what is described in Figure A.3.

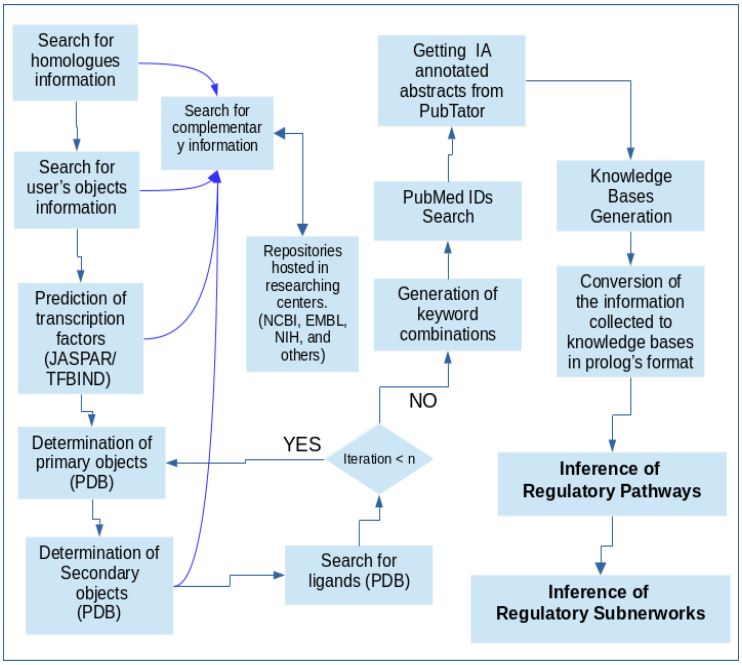

Figure A.2. Sequence of processes proposed to implement the semantic modeling of a GRN.

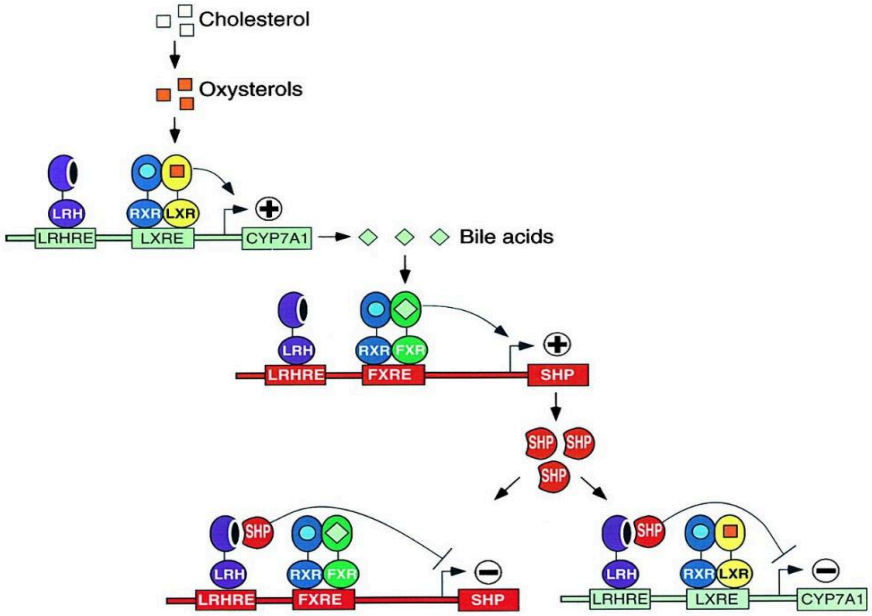

Figure A.3. A regulatory subnet of the BAXS system. Source: Lu TT, et al (2000).

1.1. A chain of processes

BioPatternsg basically consist of a set of sequentially connected programs (or processes) (see Figure A.2). Figure A.2 shows exploration levels that provide new objects accompanied by its corresponding supplementary information. Once the modeling levels indicated by the user are completed, the system proceeds to define possible PubMed IDs related to the names and synonyms for the objects in the network. Then, we use the services provided by the NCBI PubTator portal to download the abstracts for the PubMed Ids, accompanied by annotations, which include the recognition of the biological objects and the extraction of their relations. Once all the AI annotated abstracts are collected a knowledge base of regulatory events is organized (see Table A.1). Subsequently, the modeled KB is delivered to an inference system, making it possible to discover possible pathways and subnets. Such pathways should ideally show routes from the cell membrane (region close to ligands) to the nuclear membrane (region close to transcription factors) (see Figure A.1). Figure A.1 also shows that other knowledge bases are organized to support the inference processes that lead to pathways and subnets (See Table A.2)

Figure A.3 shows an example taken from the BAXS system (Lu TT et al, 2000) regarding a regulatory network. In Figure A.3 there we see two main proteins, CYP7A1 and SHP. To discover subnets like the one shown in Figure A.3, our approach is guided by determining regulatory pathways that are connected through linking events. The main idea is to determine the subnets in which the proteins of interest are stimulated and inhibited. For example, if a subnet is found in which CYP7A1 is stimulated (first) and inhibited (later), and it is possible to observe in it some other protein that mediates between both scenarios, then other subnets could be determined for such a mediating protein, which suggests the conditions that govern its stimulation or inhibition. In the example, CYP7A1 is stimulated and inhibited depending on the presence or absence of SHP, therefore it is important to find subnets in which SHP is stimulated or inhibited. Two subnets coexist in Figure A.3. One that shows the conditions that stimulate and inhibit the presence of CYP7A1, and another that describes the same for SHP. Our system offers a semi-automatic scanning option to discover scenarios like the one described above.

Table A.2. Four additional KBs for the BAXS System GRN.

1) Names, synonyms and retrieved basic facts	**2) MESH ontology**
cyp7a1_protein_human('CYP7A1'). synonyms('CYP7A1', ['cytochrome P450 family 7 subfamily A member 1', 'CYP7', 'Cytochrome P450 7A1', ...]). nr1h3_protein_mouse('NR1H3'). synonyms('NR1H3',['LXR-a', 'RLD-1','LXRa', 'Oxysterols receptor LXR-alpha', 'Liver X receptor alpha', 'Nuclear receptor subfamily 1 group H member 3', ...]). ..	object('CYP7A1'). is_a('CYP7A1','CYP7A1 protein, human'). is_a('CYP7A1 protein, human','Cholesterol 7-alpha-Hydroxylase'). is_a('Cholesterol 7-alpha-Hydroxylase','Cytochrome P450 Family 7'). is_a('Cytochrome P450 Family 7','Cytochrome P-450 Enzyme System'). is_a('Cytochrome P-450 Enzyme System','Cytochromes'). is_a('Cytochromes','Enzymes and Coenzymes'). is_a('Cytochromes','Hemeproteins'). is_a('Hemeproteins','Proteins'). is_a('Cytochrome P-450 Enzyme System','Mixed Function Oxygenases'). is_a('Mixed Function Oxygenases','Oxygenases'). is_a('Oxygenases','Oxidoreductases'). is_a('Oxidoreductases','Enzymes'). is_a('Cholesterol 7-alpha-Hydroxylase','Steroid Hydroxylases'). is_a('Steroid Hydroxylases','Cytochrome P-450 Enzyme System'). object('NR1H3'). .. ..
3) GeneOntology trees	**4) Logical facts inferred from 1, 2 and 3 (improved manually)**
mf('CYP7A1',['cholesterol 7-alpha-monooxygenase activity', 'heme binding', 'iron ion binding']). bp('CYP7A1',['bile acid biosynthetic process', 'cellular response to cholesterol', 'sterol metabolic process', ...]). cc('CYP7A1',['endoplasmic reticulum membrane', 'intracellular membrane-bounded organelle', 'organelle membrane']). is_a('cholesterol 7-alpha-monooxygenase activity', 'steroid hydroxylase activity'). is_a('steroid hydroxylase activity','monooxygenase activity'). ..	protein('CYP7A1'). enzyme('CYP7A1'). receptor('NR1H4'). protein('NR1H4'). receptor('SHP'). protein('SHP'). ligand('bile acid'). ligand('oxysterols'). ligand('cholesterol'). receptor('RXR'). protein('RXR'). receptor('NR5A2'). protein('NR5A2'). receptor('LRH'). protein('LRH'). receptor('NR1H3'). protein('NR1H3'). receptor('NR1H2'). protein('NR1H2'). transcription_factor('RUNX1'). protein('RUNX1'). transcription_factor('PBX1'). protein('PBX1'). transcription_factor('CLIP1'). protein('CLIP1'). protein('CLIP2'). ligand('CLIP3'). protein('CLIP3'). enzyme('BAAT'). transcription_factor('CAP'). ..

1.2. About the knowledge gathered.

Figures A.1 and A.2 illustrate the processes that guide the development of the various knowledge bases necessary to describe the different objects collected for a GRN. Table A.1 shows an example, corresponding to a set of events like the one illustrated in Figure A.3 (Lu TT et al, 2000). Table A.1 shows events in which a subject somehow affects an object and the syntax to represent this is: event (subject, relationship, object). The subjects and objects visible in Table A.1 illustrate

the molecular species collected in a modeling process. Considering each of those objects, BioPatternsg builds four additional KBs in order to define: 1) name, synonyms and other basic definitions, from PubMed (Fiorini et al, 2017), Protein Data Bank (PDB) (Rose et al, 2017) (Berman et al, 2014), HGNC (HUGO Gene Nomenclature Committee) (Gray et al, 2016), GeneOntology (Thomas, 2017), Uniprot (Pundir, 2017), Mesh (Baumann, 2016) and Pathway Commons (Rodchenkov et al, 2019); 2) MESH ontology, 3) GeneOntology ontology (molecular function, biological processes and cellular component); and 4) logical facts, which establish that an object satisfies the constraints that guide the analysis of pathways and subnets. The next section describes how to build those KBs and proceed in the modelling and analyzing of a GRN.

In Biopatternsg. knowledge bases like the one in Table A.1 generated with the assistance of the PubTator AI (Wei et al, 2024), shows regulatory events in which a subject affects an object in some way, and the syntax to represent this is: event(subject, relation, object). The subjects and objects visible in Table A.1 illustrate the molecular species collected and, for each one of them, our system builds four additional KBs (see Table A.2), corresponding to: 1) official name and symbol, its synonyms and other basic definitions, from PubMed (Fiorini et al, 2017), Protein Data Bank (PDB) (Rose et al, 2017) (Berman et al, 2014), HGNC (HUGO Gene Nomenclature Committee) (Gray et al, 2016), GeneOntology (Thomas, 2017), Uniprot (Pundir, 2017), Mesh (Baumann, 2016) and Pathway Commons (Rodchenkov et al, 2019); 2) MESH ontology , 3) Gene Ontology (molecular function, biological processes and cellular component); and 4) logical facts, which establish the objects in the KB of regulatory events, that satisfy the constraints that guide the analysis of signaling pathways and subnetworks. As we mentioned before, the KBs listed in 1, 2, 3 and 4 are automatically generated, but the knowledge in 4 is normally curated by an expert to improve the quality of the results. More details regarding our work can be found at Biopatternsg's github repository.

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